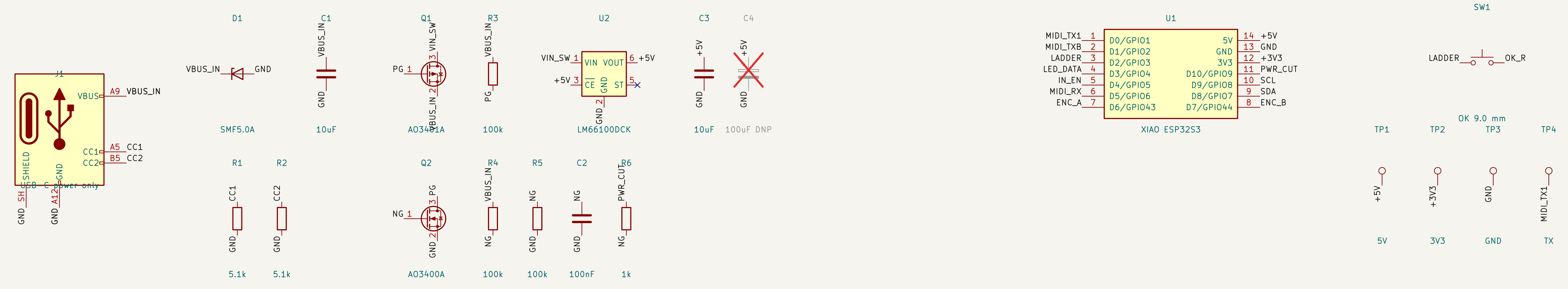
## POWER IN: J1 power only (D+/D− not fitted). Q1 cut (on by default), U2 ideal diode onto the XIAO's 5V

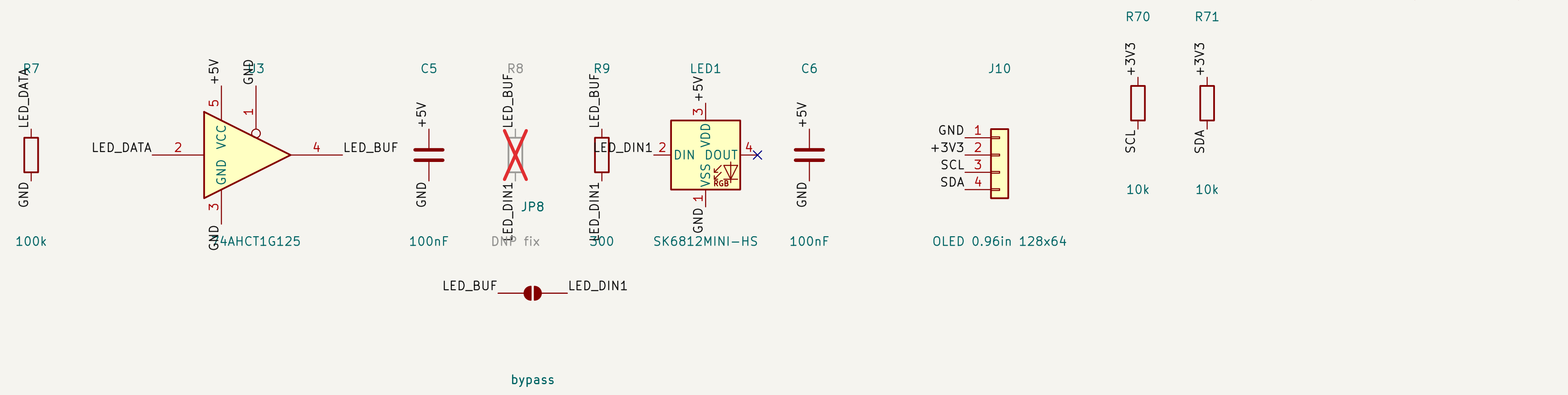
U2 ~CE tied to VOUT: the datasheet's ideal–diode mode (blocks current back into J1). The XIAO's 5V pin IS its USB–C VBUS (no part between, Seeed v1.2 schematic): a computer on the XIAO port and a charger on J1 share +5V. Firmware pulls PWR_CUT low once a computer has enumerated it (device mode): Q2 off, Q1 off, J1 cut. At reset PWR_CUT floats: R4/R5 hold Q2 on, so a charger alone always starts it.
C4 (100 uF) not fitted: host–mode hot–plug reserve, against USB's 10 uF limit for a device.

## XIAO ESP32S3 (Seeed footprint, 2 x 7 headers or castellations)

D6 = GPIO43 has 499 ohm on the module (U0TXD). D2 = GPIO3 strapping, ignored by default eFuses. ENC_A = D6 = GPIO43, ENC_B = D7 = GPIO44 (U0TXD / U0RXD), IN_EN = D4 = GPIO5, MIDI_RX = D5 = GPIO6 (v5).
v5: MIDI_TX1 = D0 = GPIO1 (OUT1 alone: Hi–Z in IN mode); MIDI_TXB = D1 = GPIO2 drives OUT2, OUT3 and OUT4 together (3 loops on one pad).
LADDER = D2 = GPIO3 (ADC1): OK / BACK / PUSH on one resistor ladder. LED_DATA = D3 = GPIO4. SDA = D8 = GPIO7, SCL = D9 = GPIO8 (the OLED and the PCF8574, I2C by the GPIO matrix, 100 kHz max for the PCF8574). PWR_CUT = D10 = GPIO9, low = J1 cut.


D1
VBUS_IN
GND
SMF5.0A
C1
10uF
Q1
AO3401A
R3
100k
U2
VIN_SW
VIN VOUT
CE GND ST
+5V
LM66100DCK
C3
10uF
C4
100uF DNP
J1
VBUS_IN
CC1
CC2
USB–C power only
R1
5.1k
R2
5.1k
Q2
AO3400A
R4
100k
R5
100k
C2
100nF
R6
1k
PWR_CUT
U1
MIDI_TX1
MIDI_TXB
LADDER
LED_DATA
IN_EN
MIDI_RX
ENC_A
D0/GPIO1
D1/GPIO2
D2/GPIO3
D3/GPIO4
D4/GPIO5
D5/GPIO6
D6/GPIO43
5V
GND
3V3
D10/GPIO9
D9/GPIO8
D8/GPIO7
D7/GPIO44
+5V
+3V3
PWR_CUT
SCL
SDA
ENC_B
XIAO ESP32S3
SW1
LADDER
OK_R
OK 9.0 mm
TP1
TP2
TP3
TP4
5V
3V3
GND
TX
MIDI_TX1

## LED: one SK6812MINI–HS on +5V (top); U3 lifts the 3.3 V data to 5 V (VIH = 0.7 x VDD = 3.5 V at 5 V). SCREEN: J10, a 0.96 in 128 x 64 OLED (SSD1315, I2C 0x3C, 3V3)

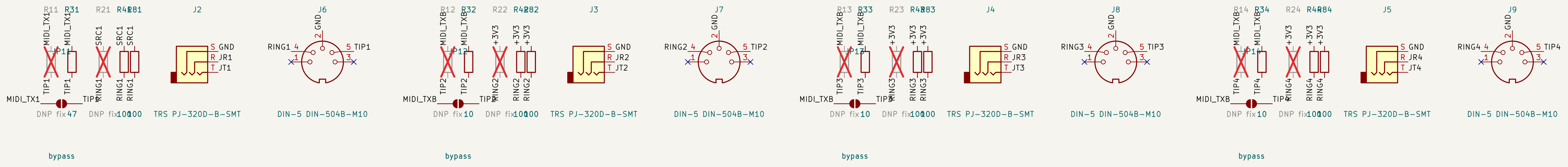

GND
VBUS_IN
VIN_SW
R7
LED_DATA
100k
U3
LED_BUF
74AHCT1G125
C5
100nF
R8
DNP fix
JP8
R9
LED_DIN1
LED1
DIN DOUT
VDD VSS
SK6812MINI–HS
C6
100nF
J10
GND
+3V3
SCL
SDA
OLED 0.96in 128x64
R70
R71
10k
10k
LED_BUF
LED_DIN1
bypass

On the board each pair is ONE footprint (J6..J9, jank:MIDI_OUT_PJ–320A_top_DIN–504B–M10_bottom: the jack's pads T, R, S carry J2..J5's nets, its switch pin SW open; the DIN's pins on the bottom). J2..J5 are BOM lines only.

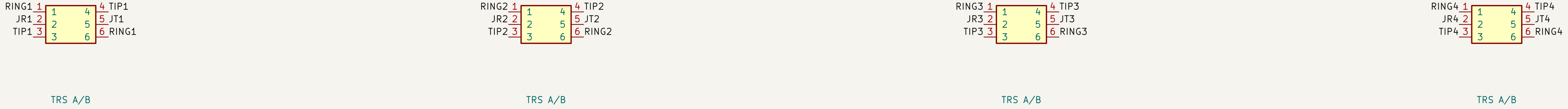
## MIDI OUT 1–4 (CA–033 at 3.3 V), v5: BOTH fitted on every output, the TRS (type A, top) AND the DIN–5 (bottom), in parallel; plug one at a time

TRS type A: tip = data (10 ohm to the TX pad), ring = source (33 ohm 0.5 W to 3V3), sleeve = GND.  DIN–5: pin 5 = tip net, pin 4 = ring net, pin 2 = GND, pins 1 and 3 open.

**OUT1: J2 TRS and J6 DIN** **OUT2: J3 TRS and J7 DIN** **OUT3: J4 TRS and J8 DIN** **OUT4: J5 TRS and J9 DIN**


R11
R31
R21
R41
R81
J2
J6
MIDI_TX1
TIP1
RING1
SRC1
DNP fix
47
100
TRS PJ–320D–B–SMT
DIN–5 DIN–504B–M10
bypass
R12
R32
R22
R42
R82
J3
J7
MIDI_TXB
TIP2
RING2
10
R13
R33
R23
R43
R83
J4
J8
TIP3
RING3
R14
R34
R24
R44
R84
J5
J9
TIP4
RING4
S GND
R JR1
T JT1

Each MIDI resistor (and R8): axial, hand–swappable; JP across it = bypass (0 ohm, open by default); R3k / R4k / R9 in parallel, not fitted (lower the value). JP2k bridged = RING on 3V3 directly (JP21: on SRC1): a mono plug then shorts the rail.

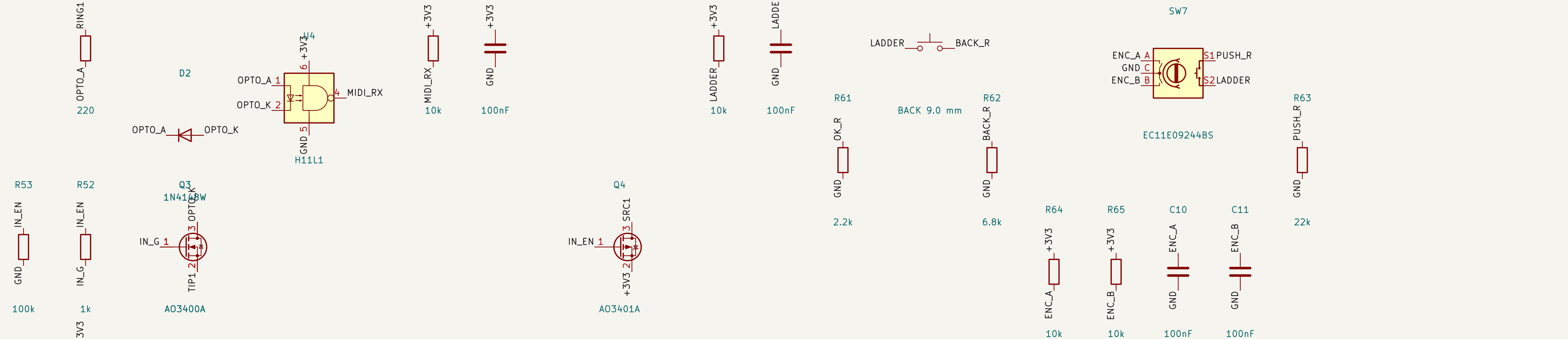

SW2
RING1
JR1
TIP1
TIP1
JT1
RING1
TRS A/B
SW3
SW4
SW5

SW2..SW5: TRS type A / B per output (top side): they swap the jack's T and R between TIPk (data, sink) and RINGk (source) after the resistors; the DIN keeps pin 5 = TIPk, pin 4 = RINGk. 1–2 / 4–5 closed = type A.

## OUT1 AS MIDI IN (software): TX1 to input (Hi–Z), IN_EN high (Q4 off: no source, Q3 on): MIDI_RX by UART RX

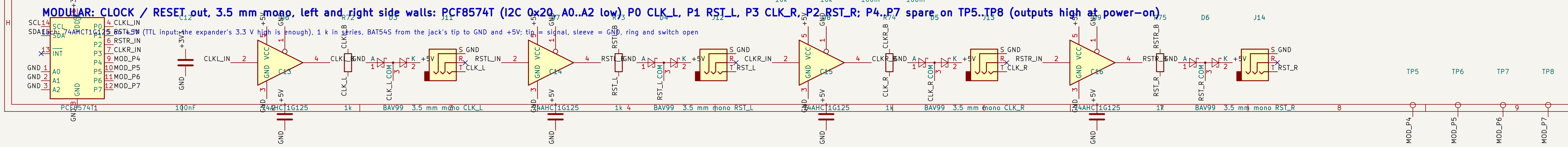
## CONTROLS (top side, by the screen): one ADC ladder on GPIO3: R60 10k to 3V3; OK 2.2k, BACK 6.8k, PUSH 22k to GND

OUT mode: IN_EN low (R53 at reset): Q4 on (3V3 to the 33 ohm), Q3 off, the opto's loop open (its body diode blocks). [illegible] Idle 3.30 V, OK 0.60 V, BACK 1.34 V, PUSH 2.27 V [illegible] Encoder A/B: 10k to 3V3, 100 nF to GND each.


R50
RING1
OPTO_A
220
D2
OPTO_A
OPTO_K
U4
MIDI_RX
H11L1
R51
10k
C8
100nF
R53
IN_EN
100k
R52
IN_G
1k
Q3
1N4148W
AO3400A
Q4
SRC1
AO3401A
R60
LADDER
10k
C9
100nF
SW6
LADDER
BACK_R
BACK 9.0 mm
R61
OK_R
2.2k
R62
BACK_R
6.8k
SW7
ENC_A
GND
ENC_B
PUSH_R
LADDER
EC11E09244BS
R63
PUSH_R
22k
R64
R65
C10
C11
ENC_A
ENC_B
10k
10k
100nF
100nF

## MODULAR: CLOCK / RESET out, 3.5 mm mono, left and right side walls: PCF8574T (I2C 0x20, A0..A2 low) P0 CLK_L, P1 RST_L, P3 CLK_R, P2 RST_R; P4..P7 spare on TP5..TP8 (outputs high at power–on)

Each: 74AHCT1G125 on +5V (TTL input: the expander's 3.3 V high is enough), 1 k in series, BAT54S from the jack's tip to GND and +5V; tip = signal, sleeve = GND, ring and switch open


U5
SCL
SDA
INT
A0
A1
A2
P0
P1
P2
P3
P4
P5
P6
P7
CLKL_IN
RSTL_IN
RSTR_IN
CLKR_IN
MOD_P4
MOD_P5
MOD_P6
MOD_P7
PCF8574T
C12
100nF
U6
C13
74AHCT1G125
R72
CLK_L
1k
D3
BAV99
J11
3.5 mm mono CLK_L
U7
C14
R73
RST_L
D4
J12
3.5 mm mono RST_L
U8
C15
R74
CLK_R
D5
J13
3.5 mm mono CLK_R
U9
C16
R75
RST_R
D6
J14
3.5 mm mono RST_R
TP5
TP6
TP7
TP8
MOD_P4
MOD_P5
MOD_P6
MOD_P7

generated by make_schematic.py; do not edit by hand
hardware/pcb/README.md: the power path, the pin table and every unverified item

Sheet: /
File: jank–pcb–mod.kicad_sch

**Title: jank: XIAO ESP32S3 MIDI clock stick**

Size: A2
Date: 2026–11–08
**Rev: 0.1 schematic for review**
KiCad E.D.A. 10.0.8
Id: 1/1